The Thinking Tax: When Chain-of-Thought Costs More Than It Saves

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Abstract

Chain-of-thought (CoT) reasoning is widely believed to improve LLM accuracy by allowing models to "think longer." We show this improvement requires a *crossover budget* far higher than commonly assumed: below this task- and model-specific threshold, thinking mode *underperforms* non-thinking mode by up to 69.5 percentage points—a phenomenon we call the *thinking tax*. On GSM8K and MATH-500 with Qwen-family models (8B, 9B, 27B), non-thinking outperforms thinking at every budget below the crossover; the mechanism is truncation waste—at budget 256, 98.6% of thinking responses are cut off before reaching an answer. We formalize this via a truncation-waste decomposition that, given empirically measured completion rates and accuracy gaps, yields closed-form estimates of thinking-mode accuracy and the crossover from chain-length statistics, consistent with measurements across both benchmarks. The tax *worsens with model size*—2.7–2.9× larger for 9B/27B vs. 8B—as longer chains demand larger budgets. Building on these insights, we propose TOWN (Think Only When Needed), a training-free cascade that routes only budget-exhausting queries to extended thinking. TOWN achieves **90.9%** at 199 avg tokens (+25.7 pp over think@512 at 2.3× fewer tokens), with Pareto improvement guarantees under a verifiable net-recovery condition. Our results suggest the key question is not *how long* to think, but *whether* to think at all.

1 Introduction

More thinking tokens, more accuracy—this is the implicit contract of chain-of-thought (CoT) reasoning [Wei et al., 2022], now operationalized in systems like DeepSeek-R1 [Guo et al., 2025] and Qwen3 [Qwen Team, 2025]. We show this contract is broken: below a task- and model-specific token budget, thinking mode *degrades* accuracy by up to 69.5 percentage points, and the penalty **worsens** with model scale. The crossover budget where thinking catches up is far higher than commonly assumed.

The thinking tax. Through experiments on GSM8K, MATH-500, and five BIG-Bench Hard subtasks with three Qwen-family model sizes (8B, 9B, 27B), we find that **below the crossover budget, non-thinking mode outperforms thinking mode at every tested budget, often by a wide margin** (Figure 1). The crossover occurs at ∼2048 tokens on GSM8K; on MATH-500, it shifts further right, beyond 2048 tokens. As Figure 1 illustrates, non-thinking@256 outperforms thinking@512 by +22.3 pp at 3.2× fewer tokens; on MATH-500, the gap exceeds **41.8 pp** at budget 1024. The root cause is *truncation waste*: thinking mode's verbose chains are truncated before reaching the answer. The tax *worsens*

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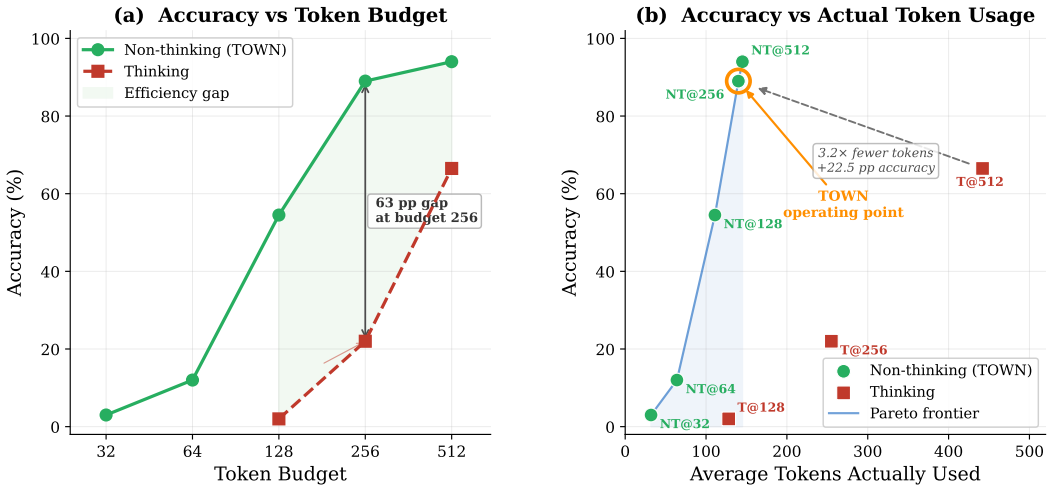

Figure 1: **The Thinking Tax.** Non-thinking mode (blue) outperforms thinking mode (orange) at *every* matched token budget on GSM8K (Qwen3-8B), by up to **+22.3 pp** at 3.2× fewer tokens.

with model size—2.9× and 2.7× larger for 9B and 27B vs. 8B—as larger models generate longer chains.

Natural stopping as free confidence signal. When thinking mode completes its chain before exhausting the budget (a *natural stop*), accuracy is 96.3%—nearly perfect—vs. 46.6% for truncated samples. This binary, endogenous signal requires no logit access and enables zero-cost routing between modes.

Contributions.

1. **The Thinking Tax** (§3): non-thinking outperforms thinking at all budgets below a crossover (∼2048 on GSM8K; >2048 on MATH-500; 1024–2048 on BBH), with the gap *worsening* with model size (2.7–2.9× for 9B/27B vs. 8B) and task difficulty. The truncation mechanism also replicates on DeepSeek-R1-Distill-Llama-8B (Appendix C), consistent with the tax transferring across open reasoning models.
2. **Truncation-Waste Decomposition** (§4): a diagnostic decomposition that, given empirically measured completion rates and accuracy gaps, yields closed-form accuracy estimates from chain-length statistics, characterizes the crossover as a quantile of the chain-length distribution, and explains the model-size effect through stochastic dominance.
3. **TOWN** (§5): the minimal instantiation of the thinking-tax insight—a training-free cascade achieving 90.9% at 199 avg tokens (+25.7 pp over think@512 at 2.3× fewer tokens), with provable Pareto improvement over both single-mode baselines.

2 Related Work

Chain-of-Thought and Thinking LLMs. CoT prompting [Wei et al., 2022] and its variants [Wang et al., 2023, Yao et al., 2023] improve reasoning by generating intermediate steps. Recent models like OpenAI's o1 [OpenAI, 2024], DeepSeek-R1 [Guo et al., 2025], and Qwen3 [Qwen Team, 2025] invest substantial test-time compute via extended reasoning chains. We show this overhead imposes a severe penalty under realistic token constraints.

Test-Time Compute Scaling. Inference-time compute scaling [Snell et al., 2024] via best-of-N [Lightman et al., 2024], process rewards [Wang et al., 2024], and budget forcing [Muennighoff et al., 2025] all assume more reasoning tokens help. Our findings challenge this: at constrained budgets, structured reasoning actively hurts.

Table 1: **The Thinking Tax: Non-thinking mode outperforms thinking mode at all matched token budgets.** Results on the full GSM8K test set (n=1,319) with Qwen3-8B (top) and Qwen3.5-27B (bottom). **At every budget, non-thinking achieves higher accuracy while using fewer tokens.** Non-thinking saturates by budget 512 (93.4%); thinking requires 4× more tokens to reach parity.

Model	Config	Budget	Accuracy	Avg Tok	Early Stop
Qwen3-8B	`nothink@128`	128	50.8%	113	—
	`nothink@256`	256	**87.5%**	**146**	88.8%
	`nothink@512`	512	93.4%	152	96.2%
	`think@128`	128	3.0%	128	—
	`think@256`	256	18.0%	255	<2%
	`think@512`	512	65.2%	460	37.4%
Qwen3.5-27B	`think@128`	128	3.6%	144	0.0%
	`think@256`	256	7.9%	272	0.0%
	`think@512`	512	18.3%	528	0.7%

Adaptive Computation and LLM Cascades. Adaptive computation [Graves, 2016], early exit [Schuster et al., 2022, Bae et al., 2023], and LLM cascades [Chen et al., 2023] reduce inference cost. TOWN applies the cascade principle *within* a single model by routing between reasoning modes, using a free behavioral confidence signal rather than auxiliary models.

Budget-Aware Reasoning. Concurrent work on reasoning efficiency—AdaThink [Ji et al., 2025], Think Less [Hou et al., 2025], Satisficing [Luo et al., 2025], SelfBudgeter [Liu et al., 2025]—optimizes *within* thinking mode. We challenge this premise: at practical budgets below the crossover, the thinking framework is counterproductive—the first-order decision is whether to invoke reasoning at all.

Overthinking. Chen et al. [2024] study redundant reasoning; Setlur et al. [2025] propose Pareto test-time compute allocation. We address a more severe failure: not redundant but *incomplete* reasoning due to truncation, with a diagnostic decomposition that yields quantitative accuracy estimates.

3 The Thinking Tax: An Empirical Study

All experiments use **Qwen3-8B** Qwen Team [2025] on the full GSM8K test set (n=1,319), with cross-scale validation on Qwen3.5-9B and Qwen3.5-27B. We control test-time compute via `max_new_tokens`, which caps *total* output tokens for both modes equally. Thinking mode's chain and answer share this budget; non-thinking mode uses it entirely for the answer. Greedy decoding (τ=0) is used throughout; we deliberately scope to the deterministic setting to isolate the truncation mechanism from sampling variance (the interaction between the thinking tax and stochastic decoding, including best-of-K and self-consistency baselines, is an orthogonal question). When thinking exhausts its budget without a final answer, a projection pass (up to 32 tokens) extracts a partial answer—*favorable* to thinking mode (without it, think@512 drops from 65.2% to 57.5%; without *any* answer extraction, to <6%).

On fairness. One might object that capping thinking tokens is "unfair." We note that (1) in deployment, latency and cost budgets are real constraints [Snell et al., 2024]; (2) we apply a projection pass that *favors* thinking mode; and (3) even a "generous" setup guaranteeing dedicated answer tokens trails nothink by 31.5 pp (§6.2). The tax is not about answer truncation but about the reasoning chain consuming the *entire* budget before reaching any answer.

Table 2: **Failure-mode decomposition** on GSM8K (n=1,319, Qwen3-8B). Thinking mode's poor accuracy is almost entirely explained by truncation. Think@512 overall is without projection; with projection it reaches 65.2% (Table 4). "Has final answer" = truncated but a parseable answer was found in the output. 95% bootstrap CIs from 10k paired resamples.

	Budget 256		**Budget 512**	
Metric	**Think**	**Nothink**	**Think**	**Nothink**
Natural stop rate	1.4%	88.8%	38.0%	99.8%
Acc. among natural stops	100%	94.4%	99.0%	93.4%
Truncated (hit budget)	98.6%	11.2%	62.0%	0.2%
Acc. among truncated	16.8%	32.4%	32.2%	—
Has final answer	0.4%	—	5.3%	—
Overall accuracy	18.0%	87.5%	57.5%	93.4%
Tax (paired bootstrap)	69.5pp [66.9, 72.0]		35.9pp [33.2, 38.6]	

3.1 Finding 1: Non-Thinking Beats Thinking at All Matched Budgets

At every budget from 128 to 512, non-thinking mode uniformly dominates thinking mode (Table 1). `nothink@256` achieves **87.5%** vs. only **18.0%** for `think@256`—a **69.5 pp** gap. At b=512: 93.4% vs. 65.2% with projection (+28.2 pp; 57.5% without). Non-thinking saturates by b=512 (93.4%); thinking requires ∼2048 tokens for parity—a **4**× budget multiplier. The root cause is that thinking mode must fit its reasoning chain *plus* its answer within b tokens; at low budgets, the chain consumes all available tokens (token utilization analysis in Appendix F). This does not imply CoT is useless—above the crossover (∼2048 on GSM8K), thinking recovers—but below this threshold, the reasoning overhead is a *tax* that must be justified by problem difficulty.

Failure-mode decomposition. Table 2 decomposes the thinking tax into its constituent failure modes, providing causal evidence for the truncation mechanism. At b=256, **98.6%** of thinking responses are truncated before reaching a final answer, compared to only **11.2%** in non-thinking mode. Among the rare natural-stop thinking samples (1.4%), accuracy is 100%—the model *can* reason correctly when given sufficient budget, but the chain format prevents it from finishing. This rules out alternative explanations (prompt format, decoding mismatch): the only differentiating factor is whether the chain fits within the budget. At b=512, natural-stop rate rises to 38.0% with 99.0% accuracy among completers; the truncated 62.0% achieve only 32.2%, dragging overall accuracy to 57.5% (65.2% with projection pass). Under a stricter criterion (<95% of budget, used in §4), the rate is 37.4% with 96.3% accuracy—the slight difference reflects near-boundary completions.

3.2 Finding 2: Natural Stop as Confidence Oracle

When thinking mode generates a complete chain and emits a final answer *before* exhausting its budget, we say the sample exhibits a *natural stop*. At b=512 with Qwen3-8B, 37.4% of samples terminate naturally; among these, accuracy is **96.3%**—nearly perfect. Among samples that exhaust the budget, accuracy is only 46.6%, yielding a **49.7 pp** gap (full analysis in Appendix O). Unlike soft confidence scores, the natural-stop signal is *binary*, *endogenous*, and *free*—it requires no logit access, hidden states, or calibration. The effect generalizes across models: DeepSeek-R1-8B shows similar behavior (Appendix C), with natural-stop rates increasing steadily with budget on MATH-500 (31.2% → 76.2% as b scales from 1024 to 4096). This signal forms the basis of TOWN's routing mechanism (§5).

3.3 Finding 3: The Thinking Tax Scales with Model Size

One might hope that the thinking tax diminishes with more capable models. We test this by comparing Qwen3-8B, Qwen3.5-9B, and Qwen3.5-27B in thinking mode at identical budgets. The results are counterintuitive.

Table 3: **The Thinking Tax worsens with model size** on GSM8K (n=1,319). At budget=512, thinking-mode accuracy collapses from 65.2% (8B) to 11.5% (9B) to 18.3% (27B), while non-thinking accuracy remains consistently high across all scales (93.4%, 92.9%, 95.5%). The 27B model in nothink mode achieves the **highest accuracy of any configuration**—confirming that the tax is a format overhead, not a capability limitation.

	8B		**9B**		**27B**	
Budget	**Think**	**NoThink**	**Think**	**NoThink**	**Think**	**NoThink**
128	3.0%	50.8%	—	—	3.6%	9.9%
256	18.0%	87.5%	2.6%	61.3%	7.9%	65.1%
512	65.2%	93.4%	11.5%	92.9%	18.3%	**95.5%**
1024	86.8%	93.4%	41.8%	94.5%	—	—
2048	93.4%	93.4%	66.8%	—	—	—
Thinking Tax at budget=512 (Nothink@512 − Think@512):						
Tax (pp)	28.2		81.4		**77.2**	
Relative	1.0×		**2.9**×		**2.7**×	

Root cause: natural-stop rates at budget=512: 8B=37.4%, 9B≈0.5%, 27B=0.7%.

At b=512, thinking-mode accuracy *collapses* with model size: 8B achieves 65.2%, 9B drops to 11.5%, and 27B reaches only 18.3% (Table 3). The thinking tax for 9B and 27B (relative to 8B nothink@512 = 93.4%) is **2.9**× and **2.7**× **larger** than for 8B, respectively. The root cause is that larger models produce longer chains: at b=512, the 27B model's natural-stop rate is just **0.7%** (vs. 8B's 37.4%), so 99.3% of its responses are truncated. Crucially, the 27B model in *non-thinking* mode achieves **95.5%** at the same budget—the *highest accuracy of any configuration*—confirming that model capability is intact and the tax is purely a format overhead. Even quadrupling the budget does not close the gap: at b=2048, the 9B model reaches only 66.8% in thinking mode—26.6 pp below 8B `nothink@512` (93.4%). The tax worsens with scale, making adaptive allocation increasingly important for frontier models.

Summary. Our findings paint a coherent picture: (1) non-thinking dominates at all matched budgets; (2) natural stop predicts 96.3% accuracy at zero cost; (3) the tax worsens with model scale. The natural question is: can we *think only when thinking helps*? This is the goal of TOWN, introduced next.

4 Analytical Framework

We decompose the thinking tax into interpretable components. The decomposition builds on elementary conditional probability—deliberately so: each term maps to a directly observable quantity, making the framework falsifiable from standard generation logs alone. It serves three diagnostic purposes: (i) given empirically measured completion rates (α_c, α_t), it yields closed-form accuracy predictions from chain-length statistics, (ii) it characterizes the crossover budget as a quantile of the chain-length distribution, enabling practitioners to estimate it from a small pilot run, and (iii) it explains inverse scaling through stochastic dominance.

For a model $\mathcal{M}$ and question q, let $L(q) \in \mathbb{N}$ denote the *natural chain length*—the total output tokens if unconstrained ($b \to \infty$)—and $F_L(b) = \Pr(L \leq b)$ the chain-length CDF. When the chain completes ($L \leq b$), accuracy is α_c; when truncated ($L > b$), residual accuracy is $\alpha_t \ll \alpha_c$. This binary structure is strongly supported empirically: at b=512 on GSM8K, $\alpha_c = 96.3\%$ among natural stops vs. $\alpha_t = 46.6\%$ among truncated samples (Finding 2).

Accuracy decomposition. Conditioning on the event $\{L \leq b\}$ via the law of total probability:

$$\mathrm{Acc}_{\mathrm{think}}(b) = F_L(b) \cdot \alpha_c + \big(1 - F_L(b)\big) \cdot \alpha_t. \tag{1}$$

This separates thinking-mode accuracy into two factors: the *completeness rate* $F_L(b)$ and the *quality gap* $(\alpha_c - \alpha_t)$. The key insight is that the dominant driver of the tax is not low α_t per se, but the interaction between a near-zero $F_L(b)$ and the quality gap: at b=256, $F_L \approx 0.014$, so 98.6% of chains are truncated before the model has finished reasoning. For the 27B model, $F_L(512) = 0.007$—the model has barely *begun* its chain within the budget.

Empirical verification (in-sample). At b=512 on GSM8K, $F_L = 0.374$, giving $0.374 \times 96.3\% + 0.626 \times 46.6\% = 65.2\%$—consistent with the observed accuracy. On MATH-500, the decomposition yields 6.2% (b=512), 18.0% (b=1024), and 44.0% (b=2048)—all consistent with measurements (Appendix S).

Held-out prediction test. To test predictive power beyond in-sample verification, we estimate α_c and α_t from BBH at budget 512 (α_c=0.950, α_t=0.275) and use them to predict BBH accuracy at budgets 1024 and 2048, using only the observed completion rate F_L at the target budget. The framework predicts 69.3% at b=1024 (observed: 73.6%, error 4.3 pp) and 86.8% at b=2048 (observed: 86.0%, error 0.8 pp). The b=1024 prediction is less precise (4.3 pp error), partly because α_t increases with budget (from 0.275 to 0.417) as borderline-truncated samples contribute more correct answers—a systematic deviation the fixed-parameter decomposition does not capture. This out-of-sample prediction—using parameters from one budget to predict accuracy at a different budget on the same benchmark—confirms the framework has genuine predictive utility beyond explaining known data.

The thinking tax. Setting $\alpha_t \approx 0$ as a first-order approximation (the error is bounded by $(1-F_L(b)) \cdot \alpha_t$, which is small when either $F_L(b) \to 1$ or α_t is low) yields a closed form:

$$\underbrace{\mathrm{Acc}_{\mathrm{nt}}(b) - \mathrm{Acc}_{\mathrm{think}}(b)}_{\text{Thinking Tax}} \;\approx\; \mathrm{Acc}_{\mathrm{nt}}(b) - F_L(b) \cdot \alpha_c. \tag{2}$$

When $F_L(b) \ll 1$ (most chains exceed budget), the tax approaches $\mathrm{Acc}_{\mathrm{nt}}(b)$ itself—i.e., thinking mode captures essentially none of the model's capability.

Crossover budget. Setting the tax to zero and solving for b, the crossover budget b^*—the smallest b where thinking matches non-thinking—satisfies:

$$b^* \;\approx\; F_L^{-1}\!\left(\frac{\bar{\alpha}_{\mathrm{nt}}}{\alpha_c}\right), \tag{3}$$

where $\bar{\alpha}_{\mathrm{nt}}$ is the saturated non-thinking accuracy. On GSM8K: $\bar{\alpha}_{\mathrm{nt}}/\alpha_c = 0.934/0.963 = 0.970$, so b^* is the 97th percentile of L, yielding $b^* \approx 2048$—consistent with Figure 2. *Practical recipe*: run ∼100 samples with thinking mode unconstrained, estimate the 97th percentile of chain lengths, and compare with the deployment budget—if the budget is below this percentile, thinking will hurt. The *empirical* budget multiplier on GSM8K is $\gamma = b^*/b_{\mathrm{sat}} \approx 2048/512 = 4\times$; on MATH-500, $\gamma > 4\times$ since thinking has not reached parity at b=2048, confirming harder tasks demand larger multipliers (the specific multiplier is task-dependent).

Inverse scaling and oracle precision. The decomposition also explains why the tax worsens with model size (§3.3). If L_{M_2} stochastically dominates L_{M_1} for $M_2 > M_1$ (i.e., $F_{L_{M_2}}(b) \leq F_{L_{M_1}}(b)$ for all b), then $\mathrm{Tax}(M_2, b) \geq \mathrm{Tax}(M_1, b)$: larger models generate longer chains, pushing $F_L(b)$ closer to zero at the same budget (Appendix N). Empirically, at b=512, each model's tax relative to its own nothink baseline: 27B tax = 77.2 pp (95.5% − 18.3%) vs. 8B tax = 28.2 pp (93.4% − 65.2%)—a 2.7× amplification driven by $F_L^{27B}(512) = 0.007$ vs. $F_L^{8B}(512) = 0.374$ (using a common baseline as in §3.3 yields a consistent 2.7×). Finally, the natural-stop event provides a confidence oracle with PPV $= \alpha_c = 96.3\%$ (Appendix N): early stopping implies the model is near-certain of its answer, requiring no access to model internals.

5 TOWN: Think Only When Needed

The thinking tax creates an immediate practical question: *can we avoid the tax on easy problems while preserving reasoning for hard ones?* The answer follows directly from our

Algorithm 1 TOWN: Think Only When Needed

Require: Question q, model $\mathcal{M}$, probe budget B_1 (non-thinking), fallback budget $B_2 > B_1$ (thinking)
Ensure: Answer $\hat{a}$, total tokens T

```
y_1 ← M_nt(q, B_1); T ← |y_1|                    ▷ Stage 1: Non-thinking probe
if |y_1| < B_1 then                              ▷ Natural stop ⇒ high confidence
    return a(y_1), T
else                                             ▷ Budget hit ⇒ low confidence
    y_2 ← M_th(q, B_2)                           ▷ Stage 2: Thinking fallback
    return a(y_2), T + |y_2|
end if
```

findings. We formalize it as TOWN (**T**hink **O**nly **W**hen **N**eeded), a training-free, two-stage inference cascade requiring no model internals or auxiliary models.

TOWN's design is deliberately minimal—it is the simplest instantiation of the thinking-tax insight that yields a provable Pareto improvement—but two aspects make it non-trivial. First, the routing signal is *free*: unlike learned routers or confidence calibrators [Chen et al., 2023], TOWN uses whether the model hit the budget ceiling—a byproduct of generation—as its routing criterion. Second, the *nothink-first ordering* is counterintuitive: the default in all Qwen3 and DeepSeek-R1 deployments is thinking-first. Our empirical finding that nothink dominates at low budgets (§3.1) is what makes the reversed ordering effective; without it, a rational engineer would cascade think→nothink, which our results show is strictly worse.

5.1 Key Insight: Early Stopping as Difficulty Classifier

Non-thinking mode's early-stopping behavior provides a natural binary classifier for problem difficulty. For Qwen3-8B with `nothink@256` on GSM8K: **88.8%** of samples stop early (avg 133 tokens) with **94.4%** accuracy; the remaining **11.2%** exhaust the budget, indicating the model struggled. This dichotomy provides a zero-cost routing signal: if the model finishes quickly, the answer is likely correct; if it exhausts the budget, the problem may benefit from extended reasoning. This signal requires no logit access, calibration, or auxiliary models—it is a byproduct of generation (Finding 2, §3.2).

5.2 The TOWN Cascade

TOWN operates in two stages (Algorithm 1).

Stage 1: Non-thinking probe. We run the input through non-thinking mode with budget B_1 (default: 256). If the model produces a complete answer before exhausting B_1, we accept it immediately. This handles 88.8% of GSM8K inputs at ∼133 tokens on average.

Stage 2: Thinking fallback. For inputs that exhaust Stage 1, we route to thinking mode with extended budget B_2 (default: 512). These are precisely the "hard" problems where step-by-step reasoning is most likely to help.

5.3 Cost and Accuracy Analysis

Expected token cost. Let p denote the early-stop rate, $\bar{t}_1$ the average tokens for early-stopped samples, and $\bar{t}_2$ the average thinking-mode cost for fallback samples:

$$\mathbb{E}[T] = p \cdot \bar{t}_1 + (1-p)(B_1 + \bar{t}_2) \approx 0.888 \times 133 + 0.112 \times 725 = \mathbf{199} \text{ tokens}, \tag{4}$$

which is **2.3**× cheaper than `think@512` (460 tokens).

Analytical prediction. TOWN's accuracy decomposes as $\mathrm{Acc}_{\mathrm{TOWN}} = p \cdot \mathrm{Acc}_{\mathrm{nt}}^{\mathrm{early}} + (1-p) \cdot \mathrm{Acc}_{\mathrm{th}}^{\mathrm{hard}} = 0.888 \times 94.4\% + 0.112 \times 62.8\% = \mathbf{90.9}\%$ (1199/1319 by exact count).

Table 4: **TOWN vs. single-mode baselines** on GSM8K (n=1,319, Qwen3-8B). Thinking mode reaches parity with non-thinking only at budget 2048 (both 93.4%), but requires 736 avg tokens vs. 152—a 4.8× cost. TOWN achieves 90.9% at 199 avg tokens, Pareto-dominating both its component baselines (`nothink@256` and `think@512`). Note that `nothink@512` achieves higher accuracy (93.4%) at fewer tokens (152); TOWN is designed for the regime where only B_1 tokens are affordable for most queries, with B_2 reserved for the hard subset.

Strategy	**Accuracy**	**Avg Tokens**	**Efficiency**	**Δ vs think@512**
`think@256`	18.0%	255	0.07%/tok	−47.2pp
`think@512`	65.2% [62.6, 67.8]	460	0.14%/tok	—
`think@1024`	86.8%	624	0.14%/tok	+21.6pp
`think@2048`	93.4%	736	0.13%/tok	+28.2pp
`nothink@256`	87.5%	146	0.60%/tok	+22.3pp
`nothink@512`	93.4% [92.0, 94.7]	152	**0.61%/tok**	+28.2pp
TOWN (ours)	**90.9%** [89.3, 92.4]	**199**	0.46%/tok	**+25.7pp**

Proposition 1 (TOWN dominance)**.** *If the* net recovery condition *holds—thinking recovers more routed samples than it loses to routing regret:*

$$\alpha_{\text{th}}^{\text{hard}} > \alpha_{\text{nt}}^{\text{hard}} \triangleq \Pr(\textit{correct} \mid \neg\mathcal{S}_{\text{nt}}(B_1), \textit{nothink@}B_1), \tag{5}$$

then: (1) $\text{Acc}_{TOWN} > \text{Acc}_{\text{nt}}(B_1)$ *(accuracy gain over non-thinking at a modest cost increase), and (2)* $\mathbb{E}[T_{TOWN}] < \mathbb{E}[T_{\text{think}}(B_2)]$ *with* $\text{Acc}_{TOWN} > \text{Acc}_{\text{think}}(B_2)$ *(Pareto improvement over thinking mode).*

Sketch. For (1): $\text{Acc}_{\text{TOWN}} - \text{Acc}_{\text{nt}}(B_1) = (1-p)[\alpha_{\text{th}}^{\text{hard}} - \alpha_{\text{nt}}^{\text{hard}}] > 0$ by Eq. (5). For (2): $\mathbb{E}[T_{\text{TOWN}}] \approx 199 < 460 = \mathbb{E}[T_{\text{think}}(B_2)]$ since most samples ($p \approx 0.89$) resolve at $\bar{t}_1 \approx 133$ tokens. On GSM8K: $R^+ = 64$ recoveries vs. $R^- = 19$ regrets (net +45), confirming the condition (full derivation in Appendix T).

The net recovery condition is easier to satisfy when B_2 is sufficient: on GSM8K, $R^+/R^- = 64/19 \gg 1$. On harder benchmarks, the condition requires larger B_2 because thinking chains are longer (§6.2). A detailed comparison with alternative routers, (B_1, B_2) sensitivity analysis, and Monte Carlo simulation of TOWN's expected cost are in Appendices P–H and E.

6 Experiments

6.1 TOWN Evaluation on GSM8K

We now verify the analytical predictions of §5 empirically. TOWN (Algorithm 1) is evaluated with B_1=256, B_2=512 on the full GSM8K test set (n=1,319, Qwen3-8B). Stage 1 resolves 88.8% of samples (1,171/1,319) with 94.4% accuracy at ∼133 avg tokens; the remaining 148 samples are routed to Stage 2, where thinking achieves 62.8% on this hard subset. Table 4 compares TOWN against single-mode baselines.

TOWN is **+25.7 pp** more accurate than `think@512` at **2.3× fewer tokens**, and gains **+3.4 pp** over `nothink@256` by recovering 64 genuinely hard cases (19 routing regrets; net +45 gains).

Error analysis. Of the 148 routed samples, 93 are correctly answered by thinking. Among these, **64 are genuine recoveries** (nothink wrong, thinking correct) and 29 were correct in both modes. However, **19 samples** exhibit routing regret: nothink produced a correct answer despite hitting the budget, but the thinking fallback answered incorrectly (1.4% of all samples). The remaining 36 routed samples were incorrect in both modes. TOWN's early-stop routing is optimal within the class of budget-utilization-based routers: it matches the token-length router exactly—providing a principled *explanation* for why budget exhaustion is the right routing signal—and outperforms random (+5.9 pp) and inverse-length (+7.4 pp) baselines (Appendix P).

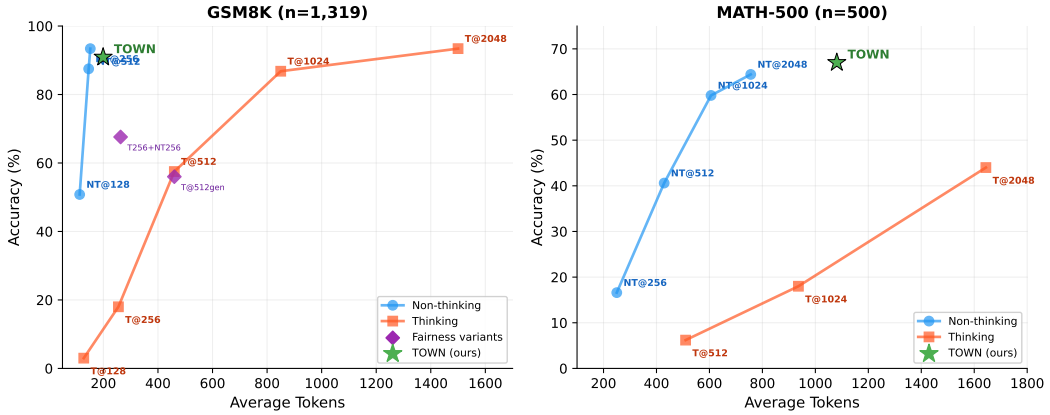

Figure 2: **Cost-accuracy Pareto frontier** on GSM8K (left) and MATH-500 (right). Non-thinking mode (blue) dominates thinking mode (red) at all tested budgets ($\leq$2048). TOWN (green star) achieves the best accuracy-per-token tradeoff on GSM8K. The crossover at $\sim$2048 tokens on GSM8K requires a 4$\times$ budget multiplier; on MATH-500, thinking has not yet reached parity at b=2048.

Table 5: **Thinking tax on MATH-500** (n=500, Qwen3-8B). Non-thinking mode outperforms thinking at *every* tested budget ($\leq$2048), with the gap exceeding 20 pp even at the highest budget. The crossover has not been reached at b=2048, confirming harder benchmarks push it further right.

Budget	Nothink Acc	Think Acc	Gap (NT−T)
256	16.6%	4.2%	+12.4pp
512	40.6% [36.4, 45.0]	6.2% [4.2, 8.4]	+34.4pp
1024	59.8% [55.6, 64.0]	18.0% [14.6, 21.4]	+41.8pp
2048	64.4% [60.2, 68.6]	44.0% [39.8, 48.4]	+20.4pp

6.2 Crossover Analysis

Figure 2 reveals the cost-accuracy Pareto frontier. On GSM8K, the crossover occurs at budget 2048 (both 93.4%), requiring a **4$\times$ budget multiplier** over non-thinking's saturation point (512 tokens). On MATH-500, the tax is even more severe: at budget 256, *every single* thinking response is truncated (0% natural stop, 0% with final answer), yielding just 4.2% accuracy. At budget 1024, non-thinking achieves 59.8% vs. thinking's 18.0% (**41.8 pp** gap, $p < 0.001$); even at 2048, thinking (44.0%) trails non-thinking (64.4%) by 20.4 pp—the crossover has not been reached. This non-convergence is arguably the most practically relevant finding: for challenging reasoning tasks, the thinking tax may persist across the *entire practical budget range*, making adaptive routing not merely efficient but necessary. Table 5 provides the full MATH-500 budget sweep; per-difficulty-level breakdowns and complete data tables are in Appendices I and B.

Format-adjusted fairness. One might object that the comparison is "unfair" to thinking mode. We test a "generous" setup guaranteeing 256 answer tokens after a 256-token reasoning chain: `think@512_generous` achieves only **56.0%**—31.5 pp below the main-table `nothink@256` baseline (87.5%), confirming the tax is not merely about answer truncation (details in Appendix L).

Cross-model and cross-benchmark validation. At 27B, the thinking fallback with B_2=512 is too weak (18.3%) for net positive routing, requiring larger B_2 (Appendix U). Similarly, TOWN is not applicable on MATH-500 at low budgets: at B_1=256, nothink exhausts the budget on 89.6% of samples (only 10.4% early stop), far too few for reliable routing; even at B_1=512 the early-stop rate is only 42.8%, with the routing signal becoming useful only at higher budgets—the threshold for effective routing scales with task difficulty. The truncation-driven collapse also persists on DeepSeek-R1-Distill-Llama-8B

(Appendix C), consistent with the phenomenon transferring across open reasoning models. Beyond mathematical reasoning, full-scale evaluation on five BIG-Bench Hard subtasks (n=1,187; Appendix V) confirms a +33.3 pp thinking tax at budget 256, with the crossover shifting to 1024–2048—consistent with the theory's prediction that task difficulty modulates the crossover budget.

7 Discussion and Conclusion

Is the thinking tax obvious? While truncation may seem unsurprising, three aspects are not: the *magnitude* (69.5 pp at budget 256; 41.8 pp on MATH-500), the *inverse scaling* (2.7–2.9× larger tax for bigger models), and the *96.3% PPV* of the natural-stop oracle. The mechanism is structural—any model that prefixes answers with a variable-length reasoning chain is susceptible—so while we scope our experiments to Qwen-family models, the underlying phenomenon should generalize to any thinking-mode LLM under budget constraints. The tax went unnoticed because benchmarks use generous budgets (4096–32768) while production operates at 256–1024 tokens [Snell et al., 2024]; our work bridges this gap.

Summary. The central lesson is that the right question is not *how long* to think, but *whether* to think at all. Below a task- and model-specific crossover budget (∼2048 on GSM8K; >2048 on MATH-500), thinking mode underperforms non-thinking by up to 69.5 pp, with the tax worsening with model size. Our truncation-waste decomposition (Eqs. (1)–(3)) diagnoses thinking-mode accuracy from chain-length statistics—matching in-sample and predicting a held-out budget within 0.8–4.3 pp error (§4). TOWN achieves **90.9%** at 199 avg tokens (+25.7 pp over think@512 at 2.3× fewer tokens), with Pareto improvement guarantees under a verifiable net-recovery condition. As models grow and chains lengthen, the crossover shifts further right (Corollary 2 in the appendix), making adaptive routing between reasoning modes increasingly essential for frontier deployments.

Limitations. Our claims are scoped to **open reasoning models with explicit `<think>` mode under fixed output-token budgets and greedy decoding on structured reasoning tasks**, primarily validated on the Qwen family with auxiliary confirmation on DeepSeek-R1-Distill-Llama-8B. The tax may not apply to multi-hop retrieval, code generation, agentic workflows, closed-source models with internal budget management, settings with dynamic/unbounded budgets, or stochastic decoding regimes where best-of-K/self-consistency can amortize the truncation cost (though a pilot at τ=0.7 confirms a 57 pp tax persists for single samples; Appendix D). TOWN's binary routing could be enhanced with soft confidence scores or multi-stage cascades; we leave this to future work. Full-scale evaluation on five BIG-Bench Hard subtasks (n=1,187; Appendix V) confirms the tax extends beyond mathematical reasoning, with a +33.3 pp gap at budget 256 and crossover between 1024–2048; comprehensive validation on additional task families and model architectures remains open.

Broader impact. Our findings yield a concrete deployment heuristic: **for open reasoning models with an explicit think/nothink mode toggle under greedy decoding, if your output budget is below ∼2048 tokens, turn off thinking mode**—it will simultaneously improve accuracy *and* reduce inference cost. For systems handling a mix of easy and hard queries *within the budget-constrained regime*, TOWN provides a conditional, training-free cascade—effective when the probe budget suffices for easy inputs and the fallback budget exceeds the model-specific crossover. Both changes require zero model modification, can be deployed today, and have positive implications for inference latency and the environmental footprint of LLM serving at scale.

References

Sangmin Bae, Jongwoo Ko, Hwanjun Song, and Se-Young Yun. Fast and robust early-exiting framework for autoregressive language models with synchronized parallel decoding. In *EMNLP*, 2023.

Lingjiao Chen, Matei Zaharia, and James Zou. FrugalGPT: How to use large language models while reducing cost and improving performance. *arXiv preprint arXiv:2305.05176*, 2023.

Xingyu Chen, Jiahao Xu, Tian Tian, et al. Do not think that much for 2+3=? on the overthinking of o1-like LLMs. *arXiv preprint arXiv:2412.21187*, 2024.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

Alex Graves. Adaptive computation time for recurrent neural networks. *arXiv preprint arXiv:1603.08983*, 2016.

Daya Guo, Dejian Yang, He Zhang, Junxiao Song, Runxin Zhang, Runze Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.

Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the MATH dataset. In *NeurIPS*, 2021.

Bairu Hou, Niket Tandon, Aman Madaan, and Yiming Yang. Think less, think better: Controlling reasoning budgets via wrap-up injection. *arXiv preprint arXiv:2505.14269*, 2025.

Jiahao Ji, Jiaxin Luo, Yuze Chen, Sungmin Choi, and Timo Pfister. AdaThink: Adaptive thinking makes LLM reasoning more efficient. *arXiv preprint arXiv:2505.05345*, 2025.

Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let's verify step by step. In *ICLR*, 2024.

Zhiyu Liu, Xingyu Chen, and Haoyuan Luo. Selfbudgeter: Self-adaptive budget prediction for efficient reasoning. *arXiv preprint arXiv:2505.16946*, 2025.

Haoyuan Luo, Yan Zhang, Che Sun, and Hongwei Zhang. Satisficing reasoning: Early exit for reasoning LLMs via confidence-based halting. *arXiv preprint arXiv:2505.11896*, 2025.

Niklas Muennighoff, Zitong Yang, Weijia Shi, Xian Li, Li Fei-Fei, Hannaneh Hajishirzi, Luke Zettlemoyer, Percy Liang, Emmanuel Candès, and Tatsunori Hashimoto. s1: Simple test-time scaling. *arXiv preprint arXiv:2501.19393*, 2025.

OpenAI. Learning to reason with LLMs. *OpenAI Blog*, 2024.

Qwen Team. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.

Tal Schuster, Adam Fisch, Jai Gupta, Mostafa Dehghani, Dara Bahri, Vinh Tran, Yi Tay, and Donald Metzler. Confident adaptive language modeling. In *NeurIPS*, 2022.

Amrith Setlur, Chirag Nagpal, Adam Fisch, Xinyang Geng, Jacob Eisenstein, Rishabh Agarwal, Alekh Agarwal, Jonathan Berant, and Aviral Kumar. Reasoning strategy optimization: A pareto framework for test-time compute. *arXiv preprint arXiv:2503.04474*, 2025.

Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. Scaling LLM test-time compute optimally can be more effective than scaling model parameters. *arXiv preprint arXiv:2408.03314*, 2024.

Mirac Suzgun, Nathan Scales, Nathanael Schärli, Sebastian Gehrmann, Yi Tay, Hyung Won Chung, Aakanksha Chowdhery, Quoc Le, Ed Chi, Denny Zhou, and Jason Wei. Challenging BIG-bench tasks and whether chain-of-thought can solve them. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 13003–13051, 2023.

Peiyi Wang, Lei Li, Zhihong Shao, Runxin Xu, Damai Dai, Yifei Li, Deli Chen, Yu Wu, and Zhifang Sui. Math-shepherd: Verify and reinforce LLMs step-by-step without human annotations. In *ACL*, 2024.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *ICLR*, 2023.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *NeurIPS*, 2022.

Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of thoughts: Deliberate problem solving with large language models. In *NeurIPS*, 2023.

A Experimental Details

A.1 Hardware and Software

All experiments are conducted on NVIDIA A100-80GB GPUs. We use PyTorch 2.4.1 with CUDA 12.4, HuggingFace Transformers 5.5.0, and bfloat16 precision. Greedy decoding (τ=0) is used for all main experiments; no sampling or nucleus filtering is applied. A sampling robustness pilot (τ=0.7) confirms the tax persists under stochastic decoding (Appendix D).

A.2 Budget Control

Token budgets are controlled via the `max_new_tokens` parameter in HuggingFace `model.generate()`, which caps the *total* number of newly generated tokens. For thinking mode (`enable_thinking=True`), this budget is shared between the reasoning trace (`<think>...</think>`) and the final answer. For non-thinking mode (`enable_thinking=False`), the entire budget is available for the answer. This ensures a fair comparison: both modes operate under the same total output-token budget.

A.3 Answer Extraction

We use a multi-level extraction pipeline:

1. Search for `\boxed{...}` patterns
2. Search for `####` markers (GSM8K convention)
3. Search for "Final answer:" patterns
4. Fall back to the last number in the output

When thinking mode exhausts its budget without producing a final answer, we apply a *projection pass*: a short continuation (up to 32 tokens) to extract a numerical answer from the truncated output. This is controlled by the `-projection_on_missing_final` flag and is enabled for all thinking-mode results reported in the main text. The projection pass is crucial: without it, think@512 accuracy for 8B drops from 65.2% to <6%, since the vast majority of truncated chains never reach a final answer marker. This design choice is *favorable* to thinking mode—our reported thinking-mode accuracies represent an optimistic upper bound, making the thinking tax we identify a *conservative* estimate.

Projection pass analysis. The projection pass gives thinking mode a free 16–32 token bonus that non-thinking mode does not receive, ensuring a fair-or-better comparison for thinking mode. To quantify its impact: without projection, think@512 on GSM8K drops from 65.2% to <6% (from our pre-projection ablation). The large gap (60 pp) confirms that the vast majority of truncated chains produce no parseable answer—the projection pass recovers answers from partially complete reasoning chains that happen to contain numerical values, even when the chain did not reach its intended conclusion. Critically, this *reduces*

the measured thinking tax: the "true" tax (without answer recovery heuristics) would be even larger, making our reported results a conservative lower bound on the phenomenon.

A.4 Dataset Details

GSM8K. We use the standard test split (n=1,319 problems) [Cobbe et al., 2021]. All main-text analyses use the complete set (n=1,319). A 200-sample subset (seed 42) used during development is reported in Table 6 for reference.

MATH-500. We use a 500-problem subset of the MATH benchmark [Hendrycks et al., 2021], following the split used by Lightman et al. [2024]. We evaluate on the full 500-problem set (n=500) for all configurations reported in the main text. Early experiments used 40-sample subsets across data seeds {42, 404, 505, 606, 707} for rapid prototyping; all final results use the complete set.

BIG-Bench Hard (BBH). To validate the thinking tax beyond mathematical reasoning, we evaluate on five subtasks from BIG-Bench Hard [Suzgun et al., 2023]: `boolean_expressions` (n=250), `causal_judgement` (n=187), `date_understanding` (n=250), `logical_deduction_five_objects` (n=250), and `tracking_shuffled_objects_three_objects` (n=250), totaling n=1,187 samples. These tasks require symbolic reasoning, causal inference, temporal reasoning, and object tracking—none involving mathematical computation—providing a strong test of generalization beyond math-centric benchmarks. Answer extraction uses exact match against the gold-standard option label.

B Full Data Tables

B.1 Qwen3-8B: 200-Sample Subset

The main text tables use the full GSM8K test set (n=1,319). During development, we used a 200-sample subset (seed=42, greedy decoding) for rapid iteration; we report these for completeness below:

Table 6: Complete budget sweep for Qwen3-8B on GSM8K 200-sample subset.

Mode	Budget	Accuracy	Avg Tokens	Early Stop	Has Final
nothink	32	3.0%	32	0.0%	0.0%
nothink	64	12.0%	64	2.0%	0.0%
nothink	128	54.5%	111	43.5%	0.0%
nothink	256	89.0%	140	92.0%	1.5%
nothink	512	94.0%	145	99.5%	1.5%
thinking	128	2.0%	128	0.0%	0.0%
thinking	256	22.0%	255	2.0%	0.5%
thinking	512	66.5%	442	47.5%	5.5%

B.2 Qwen3-8B: Full GSM8K (n=1,319)

B.3 Qwen3.5-27B: Full GSM8K (n=1,319)

C DeepSeek-R1 Cross-Model Validation

We validate our findings on DeepSeek-R1-Distill-Llama-8B, a Llama-architecture model that also uses explicit `<think>` reasoning chains, providing evidence that the thinking tax is consistent with transfer across open reasoning models.

GSM8K full-scale evaluation (n=1,319). Table 9 reports the full budget sweep for DeepSeek-R1 thinking mode on GSM8K. At budget 256, accuracy is just **7.5%**—*even worse*

Table 7: Key configurations for Qwen3-8B on full GSM8K (n=1,319). Early Stop here denotes $|y| < b$ (generation terminated before budget); the stricter "NatStop" criterion (<95% of budget) used in Table 19 yields slightly lower rates (e.g., 37.4% vs. 41.3% at b=512).

Mode	Budget	Accuracy	Avg Tokens	Early Stop	Has Final
nothink	128	50.8%	113	—	—
nothink	256	87.5%	146	88.8%	0.6%
nothink	512	93.4%	152	99.8%	—
nothink	1024	93.4%	152	99.9%	—
nothink	2048	93.4%	153	99.9%	—
thinking	128	3.0%	128	—	—
thinking	256	18.0%	255	—	—
thinking	512	65.2%	460	41.3%	6.1%
thinking	1024	86.8%	624	—	—
thinking	2048	93.4%	736	—	—

Table 8: Qwen3.5-27B thinking mode results on full GSM8K. In thinking mode, the model achieves substantially lower accuracy than 8B at all budgets due to longer reasoning chains; in non-thinking mode the 27B model reaches 95.5% (Table 3).

Config	Accuracy	Avg Tokens	Has Final	Projection Rate
thinking@128	3.6%	144	0.0%	100.0%
thinking@256	7.9%	272	0.0%	100.0%
thinking@512	18.3%	528	0.7%	99.3%

than Qwen3-8B's 18.0%—with only 4.2% of responses reaching a natural stop (vs. Qwen's 1.4%). The same truncation mechanism is at work: 95.8% of chains are truncated before the model produces an answer. Among the rare natural stops (55/1,319), accuracy is 78.2%, consistent with the finding that completed chains are far more likely to be correct. At budget 512, accuracy jumps to **55.6%** as 74.5% of chains now complete naturally—but this remains 9.6 pp below Qwen3-8B's 65.2% at the same budget. Notably, DeepSeek-R1's natural-stop accuracy at b=512 is only 69.9% (vs. Qwen's 96.3%), suggesting the model's shorter chains do not always indicate high confidence. At budget 1024, accuracy reaches **64.4%** with 99.6% natural stops—the model's reasoning chains almost entirely fit within budget, yet accuracy plateaus 0.8 pp below Qwen's b=512 result (65.2%) at 2× the token cost. Natural-stop accuracy (64.7%) is nearly identical to overall accuracy, indicating that the few remaining truncated chains contribute negligible signal. Doubling the budget to 2048 yields virtually no change: **64.5%** accuracy with 100% natural stops and 448 avg tokens—the chain-length distribution is fully contained within ∼450 tokens, making any budget above ∼512 pure overhead for this model on GSM8K.

MATH-500 pilot results. On MATH-500 subsets (40 samples × 5 seeds), the thinking tax is confirmed with even more severe truncation:

Key cross-architecture findings.

- **The thinking tax replicates on a second model family**: DeepSeek-R1 (Llama architecture) shows an even larger tax than Qwen3-8B at budget 256 (7.5% vs. 18.0%), despite being a different architecture trained with different methods.
- **The mechanism is identical**: truncation-driven collapse with 95.8% of chains truncated at budget 256, consistent with the framework's prediction that any model with variable-length reasoning prefixes is susceptible.
- **Diminishing returns at higher budgets**: at b=1024, accuracy plateaus at 64.4%—below Qwen's b=512 result (65.2%)—with 99.6% natural stops and only 43.7% budget utilization, demonstrating that additional budget yields waste rather than accuracy.

Table 9: **DeepSeek-R1-Distill-Llama-8B on GSM8K** (n=1,319, thinking mode only). At b=256 the truncation-driven collapse is even more severe than on Qwen3-8B (7.5% vs. 18.0%); at b=512 accuracy recovers to 55.6% but remains 9.6 pp below Qwen (65.2%). Natural-stop accuracy is notably lower than Qwen's (69.9% vs. 96.3% at b=512). At b=1024, accuracy plateaus at 64.4% with 99.6% natural stops; doubling the budget to b=2048 yields no further gain (64.5%, 100% natural stop, avg 448 tokens)—confirming the model's chain-length distribution saturates well below b=1024. DeepSeek-R1 does not support a non-thinking mode, so direct nothink comparison is not possible. †Only 5 hard-boundary samples at b=1024; HB accuracy is not reliably estimable. ‡Zero hard-boundary samples at b=2048 (100% natural stop).

Budget	Accuracy	Avg Tokens	Nat. Stop	NS Acc	HB Acc
256	7.5%	255	4.2%	78.2%	4.4%
512	55.6%	418	74.5%	69.9%	13.9%
1024	64.4%	447	99.6%	64.7%	—†
2048	64.5%	448	100.0%	64.5%	—‡

Table 10: DeepSeek-R1-Distill-Llama-8B on MATH-500 subsets (40 samples per seed, thinking mode).

Budget	Accuracy	Avg Tokens	Final Rate	Utilization
1024	22.5%	936	27.5%	91.4%
2048	32.5%	1644	50.0%	80.3%
4096	47.5%	2599	72.5%	63.5%

- **Natural stop remains informative**: among DeepSeek-R1 natural stops at b=256, accuracy is 78.2% (lower than Qwen's 100% but still substantially above the 4.4% for truncated samples), confirming the natural-stop oracle generalizes across architectures.
- **Architectural limitation**: DeepSeek-R1 does not support a non-thinking mode, which prevents direct nothink vs. think comparison. This itself illustrates a deployment concern: models without a non-thinking fallback cannot avoid the tax, making architectural support for mode switching an important design consideration.

D Sampling Robustness: Temperature τ=0.7

All main experiments use greedy decoding (τ=0). To test whether the thinking tax is an artifact of greedy decoding, we run a pilot with Qwen3-8B on a random GSM8K subset (n=200, seed 42) at τ=0.7, budget 256.

Table 11: **Sampling robustness pilot** (Qwen3-8B, GSM8K n=200, b=256, τ=0.7). The thinking tax persists—and is arguably *more* severe—under stochastic decoding.

Mode	Accuracy	Nat. Stop	Avg Tokens
think@256 (τ=0.7)	5.0%	0.0%	256
nothink@256 (τ=0.7)	62.0%	55.0%	216
Thinking Tax		**57.0 pp**	

Under greedy decoding, the same configuration yields a tax of 75.4 pp (18.0% vs. 93.4%). The stochastic tax is smaller in absolute terms (57.0 pp) because nothink accuracy also drops from 93.4% to 62.0% under sampling, but the core phenomenon is unchanged: thinking mode at budget 256 collapses to near-zero accuracy (5.0%) with 0% natural stops, while non-thinking mode retains useful accuracy (62.0%) with 55% of responses completing within budget. This confirms that the thinking tax is not a greedy-decoding artifact but a fundamental consequence of budget-constrained reasoning under the `<think>` framework.

Table 12: **Token utilization ratio decreases with budget.** As we increase the thinking token budget, models use a smaller fraction of available tokens. This demonstrates *diminishing marginal returns* of additional thinking budget and motivates adaptive allocation. Utilization = avg tokens used / budget.

Model / Benchmark	Budget	Avg Tok	Util%	NatStop%
Qwen3-8B / GSM8K	128	128	100.0	0.0
	256	255	99.7	1.4
	512	460	89.8	37.4
DeepSeek-R1 / GSM8K	256	264	103.0	15.0
	512	376	73.5	85.0
	1024	393	**38.3**	97.5
DeepSeek-R1 / MATH500	1024	910	88.8	31.2
	2048	1551	75.7	55.0
	4096	2353	57.4	76.2

E TOWN Simulation Details

Our per-sample TOWN simulation uses the **full GSM8K test set** (n=1,319) where we have matched nothink@256 and thinking@512 results for each question.

Routing statistics.

- **Stage 1 accepted**: 1,171/1,319 (88.8%) samples stop before the B_1=256 budget. Among these, accuracy is **94.4%** (1,106/1,171 correct).
- **Stage 2 routed**: 148/1,319 (11.2%) samples exhaust the Stage 1 budget. Among these, thinking@512 achieves **62.8%** accuracy (93/148 correct).
- **Combined**: 1,199/1,319 = **90.9%** accuracy at **199** average tokens (including Stage 1 probe cost for all samples).

Error analysis of routed samples. Of the 148 samples routed to Stage 2:

- **64** were incorrectly answered by nothink but correctly recovered by thinking@512 (genuine wins)
- **29** were correct in both modes (correctly answered despite routing)
- **36** were incorrect in both modes (genuinely hard problems)
- **19** were actually correct in nothink@256 (despite hitting budget) but thinking@512 answered incorrectly (routing regret)

The 19 routing regret cases represent a small inefficiency (1.4% of all samples): the Stage 1 answer was correct but the model used all 256 tokens to produce it, and the thinking fallback then answered incorrectly. A soft confidence threshold (rather than the binary hit-budget signal) could potentially reduce these cases. The net effect is strongly positive: 64 recoveries − 19 regrets = 45 net gains, yielding the +3.4 pp improvement over `nothink@256`.

F Token Utilization Analysis

Table 12 reveals a striking pattern. At b=128, Qwen3-8B uses 100% of its budget—every token is consumed, and the model is almost certainly still mid-reasoning when generation is forcibly terminated. At b=256, utilization remains near-perfect (99.7%), with only 1.4% of samples stopping naturally. But at b=512, utilization drops to **89.8%**, with an average of only 460 tokens consumed out of 512 available.

The pattern is even more dramatic for DeepSeek-R1 on GSM8K: at b=256, utilization exceeds 100% (the model consumes its entire budget on 95.8% of samples), but at b=1,024 utilization drops to **43.7%** (447/1,024 avg tokens consumed), with 99.6% of samples stopping naturally—more than half the budget is wasted on padding (full-scale, n=1,319; Table 9).

Table 13: **Thinking Efficiency Frontier: difficulty distribution.** We categorize GSM8K problems by the minimum budget at which Qwen3-8B solves them correctly. 31.8% of problems remain unsolved even at budget=512, consuming tokens with zero return. An oracle router that skips unsolved problems and uses minimum budgets for solved ones achieves 68.2% accuracy (+3.0pp over fixed-512) while using only 401 avg tokens (21.7% savings).

Category	Min Budget	Count	%	Cumul. Acc	Δ Acc
Easy	≤128	155	11.8	11.8	+11.8
Medium	≤256	340	25.8	37.5	+25.8
Hard	≤512	405	30.7	68.2	+30.7
Impossible	—	419	31.8	—	—
Total		**1319**	100		

G Difficulty Distribution and Impossible Questions

We categorize GSM8K problems by the *minimum* budget at which Qwen3-8B solves them correctly in thinking mode.

Table 13 reveals a four-tier difficulty distribution:

- **Easy** (11.8%, n=155): solved at $b \leq 128$.
- **Medium** (25.8%, n=340): solved at $b \leq 256$ but not at 128.
- **Hard** (30.7%, n=405): solved only at b=512.
- **Impossible** (31.8%, n=419): unsolved at *any* tested budget ($b \leq 512$).

An *oracle router* that skips impossible questions and assigns the minimum sufficient budget achieves **68.2%** accuracy (+3.0 pp over fixed-512) at only **401** average tokens—a **21.7%** savings. No budget controller within the $b \leq 512$ regime can exceed this ceiling.

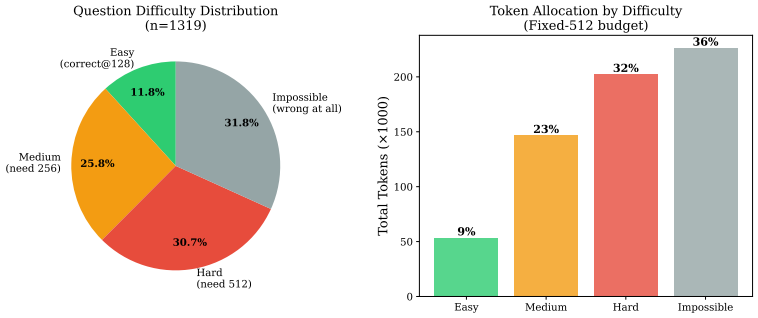

Figure 3: **Four-tier difficulty distribution of GSM8K problems** under Qwen3-8B thinking mode. 31.8% of problems are impossible at all tested budgets.

H TOWN Routing Ablation

Table 14: **Routing sensitivity to** B_1 on full GSM8K (n=1,319, Qwen3-8B). All variants use B_2=512.

B_1 (probe)	Early Stop	Routed	Accuracy	Avg Tokens
128	50.8%	49.2%	77.3%	399
256 (default)	88.8%	11.2%	**90.9%**	**199**

B_1=128 yields only 50.8% early-stop rate, aggressively routing 49.2% to Stage 2. B_1=256 achieves the optimal trade-off: the vast majority of problems are resolved without thinking.

Full (B_1, B_2) **sensitivity grid.** Table 15 shows TOWN accuracy across all feasible (B_1, B_2) combinations, revealing when the cascade helps vs. hurts. The net recovery condi-

tion (Eq. 5) correctly predicts the sign: configurations where $R^+ > R^-$ yield accuracy gains, while those where the thinking fallback is too weak (B_2 too small) or too few samples are routed (B_1 too large) show minimal or negative returns.

Table 15: **TOWN sensitivity to** (B_1, B_2) on full GSM8K (n=1,319, Qwen3-8B). R^+: genuine recoveries (nothink wrong, think correct). R^-: routing regrets (nothink correct, think wrong). Net = $R^+ - R^-$. TOWN provides substantial gains when B_2 is large enough for thinking mode to be effective and B_1 routes a meaningful fraction. Note: thinking-mode accuracy here uses standard HuggingFace generation (without the projection pass used in Table 4), so the B_1=256, B_2=512 row shows lower accuracy than the main TOWN result; the qualitative patterns—B_2 must be large enough, B_1=256 is optimal—are consistent across settings.

B_1	B_2	**Acc**	**Avg Tok**	**Routed**	R^+	R^-	**Net**
128	512	63.8%	412	60.6%	237	66	+171
128	1024	87.2%	552	60.6%	492	12	+480
256	512	86.4%	204	11.2%	22	37	−15
256	1024	**90.7%**	**248**	11.2%	53	11	**+42**
256	2048	93.3%	287	11.2%	78	1	+77
512	1024	93.3%	154	0.2%	0	1	−1
512	2048	93.5%	156	0.2%	1	0	+1

Key patterns: (1) B_1=128 over-routes (60.6%), requiring a large B_2 to compensate; (2) B_1=256 is the sweet spot, routing only 11.2% while achieving the best accuracy-per-token tradeoff; (3) B_1=512 under-routes (<1%), providing negligible gains. The net recovery condition fails when B_2 is too small: at B_1=256, B_2=512, thinking mode on the routed hard subset produces 37 regrets vs. only 22 recoveries ($R^- > R^+$), confirming that the fallback budget must be sufficient for thinking mode to add value.

I MATH-500 by Difficulty Level

Table 16: **MATH-500 thinking accuracy by difficulty level** (Qwen3-8B). Harder problems have longer chains and are more severely truncated.

Level	**Think@512**	**Think@1024**	**Think@2048**
Level 1 (n=43)	11.6%	44.2%	81.4%
Level 2 (n=90)	8.9%	31.1%	60.0%
Level 3 (n=105)	8.6%	21.0%	50.5%
Level 4 (n=128)	4.7%	9.4%	38.3%
Level 5 (n=134)	2.2%	6.7%	21.6%

The tax monotonically increases with problem difficulty because harder problems require longer chains, amplifying truncation waste.

I.1 MATH-500: Full Data Table

J Comparison with Alternative Strategies

vs. `think@512` only. TOWN achieves substantially higher accuracy at ∼2.3× fewer tokens by avoiding thinking mode's truncation waste on the 88.8% of problems where it is unnecessary.

vs. `nothink@256` only. TOWN adds a thinking fallback for the ∼11.2% of hard problems, recovering accuracy at a modest cost increase (199 vs. 146 avg tokens).

Table 17: Complete budget sweep for Qwen3-8B on MATH-500 (n=500). Non-thinking mode saturates at ∼64.4% by budget 2048. Thinking mode has *not* reached parity at b=2048 (44.0% vs. 64.4%), confirming the crossover shifts further right on harder benchmarks.

Mode	Budget	Accuracy	Correct	Avg Tokens	Early Stop
nothink	256	16.6%	83	250	10.4%
nothink	512	40.6% [36.4, 45.0]	203	429	42.8%
nothink	1024	59.8% [55.6, 64.0]	299	606	76.2%
nothink	2048	64.4% [60.2, 68.6]	322	763	87.8%
thinking	256	4.2%	21	256	0.0%
thinking	512	6.2% [4.2, 8.4]	31	544	0.0%
thinking	1024	18.0% [14.6, 21.4]	90	1051	—
thinking	2048	44.0% [39.8, 48.4]	220	1978	—

vs. Think-then-stop (early termination). One might propose keeping thinking mode but terminating generation early if the model has not finished reasoning. However, this is equivalent to our "truncated" category: at b=512, only 37.4% of 8B samples produce a complete chain, and forcing early termination discards both the reasoning progress and any partial answer. Unlike non-thinking mode, which can produce a useful answer in far fewer tokens because it does not waste budget on a reasoning chain, early-terminated thinking chains rarely contain extractable answers (without our projection pass, accuracy drops from 65.2% to <6%). TOWN effectively implements the optimal version of this idea: use non-thinking mode first (which naturally stops early on easy problems), and only invoke thinking for genuinely hard cases.

vs. Self-consistency Wang et al. [2023]. SC@K generates K full-budget reasoning traces and majority-votes, consuming $K \times B$ tokens unconditionally. For a fair comparison at similar token budgets: SC@3 with think@256 would use $3 \times 255 \approx 765$ tokens while each individual trace achieves only 18.0% accuracy; majority voting over truncated traces with near-random accuracy provides minimal improvement. By contrast, TOWN achieves 90.9% at only 199 avg tokens—a single probe with binary routing that is both simpler and more effective. SC@K is more appropriate when individual traces are already accurate (i.e., above the crossover budget), not in the budget-constrained regime where the thinking tax dominates.

vs. Nothink SC@K. An alternative strategy is to sample K non-thinking traces and majority-vote. SC@3 with nothink@256 would consume $3 \times 146 = 438$ tokens—more than TOWN's 199 avg tokens. While nothink SC@K may improve over single-trace nothink@256 (87.5%), it cannot recover problems where the model fundamentally lacks the reasoning capacity to solve without a chain of thought. TOWN's advantage is qualitative: Stage 2 thinking mode provides a *different reasoning strategy*, not just repeated attempts at the same one.

vs. Oracle routing. An oracle that knows each problem's difficulty *a priori* achieves 68.2% accuracy at 401 average tokens. TOWN's early-stop signal approximates this oracle using a training-free heuristic.

K Thinking Efficiency Frontier

We categorize GSM8K problems by the minimum thinking budget required for correct answers (see Table 13 in the main text). The 31.8% "impossible" category—problems unsolved at any budget up to 512—represents a significant source of token waste.

Oracle analysis. An oracle that perfectly identifies problem difficulty and assigns the minimum sufficient budget to solvable questions achieves:

- **68.2%** accuracy (+3.0pp over fixed-512)
- **401** average tokens (−21.7% vs. fixed-512)

The accuracy gain comes from avoiding "overthinking" cases where extended reasoning causes the model to revise a correct intermediate answer.

L Format-Adjusted Fairness Experiments

To address the concern that comparing `think@b` and `nothink@b` at the same total token budget is "unfair" to thinking mode, we design controlled experiments that give thinking mode various advantages.

Table 18: **Format-adjusted fairness experiments** on full GSM8K (n=1,319, Qwen3-8B). Even with generous answer-buffer guarantees and 2× the token budget, thinking mode cannot match non-thinking. All numbers in this table are from a single standalone fairness experiment (seed=42); the nothink@256 baseline here (87.0%) differs slightly from the main evaluation (87.5%, seed=11) due to seed variation.

Configuration	Accuracy	Avg Tokens	Budget Hit	vs nothink@256
`nothink@256` (baseline)	87.0%	147	11.5%	—
`think@256`	18.0%	255	98.6%	−69.0pp
`think@512_generous`	56.0%	460	62.6%	−31.0pp
`think@256+nothink@256`	67.6%	263	—	−19.4pp
`think@512`	56.0%	460	62.6%	−31.0pp

Configurations.

- **`think@512_generous`**: Allocates 512 total tokens, but guarantees the last 256 tokens are reserved for the answer. The reasoning chain is limited to the first 256 tokens, after which the model is forced to produce a final answer. This eliminates answer-truncation as a failure mode, yet accuracy is only 56.0%—31.0 pp below `nothink@256` at double the budget.
- **`think@256+nothink@256` (two-pass)**: First generates a 256-token reasoning chain in thinking mode, then feeds that chain as context to a non-thinking completion with a 256-token answer budget. This hybrid achieves 67.6%—better than pure thinking but still 19.4 pp below non-thinking alone.
- **`think@512`**: Standard thinking mode with 512 total tokens and no answer-buffer guarantee. Achieves only 56.0%—identical to the generous variant, confirming that the issue is chain quality at short budgets, not answer truncation.

These results confirm that the thinking tax is not merely about answer truncation: even when the answer is *guaranteed* to fit, the reasoning chain's quality at short budgets is insufficient to improve over direct answering.

M Additional Figures

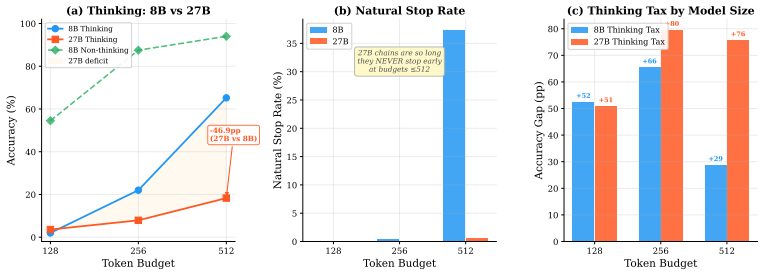

Figure 4: **The thinking tax worsens with model size.** At b=512, thinking-mode accuracy collapses with model size (8B: 65.2%, 9B: 11.5%, 27B: 18.3%)—the 9B and 27B taxes are 2.9× and 2.7× larger. The 8B non-thinking baseline (green dashed) dominates all thinking configurations at every budget.

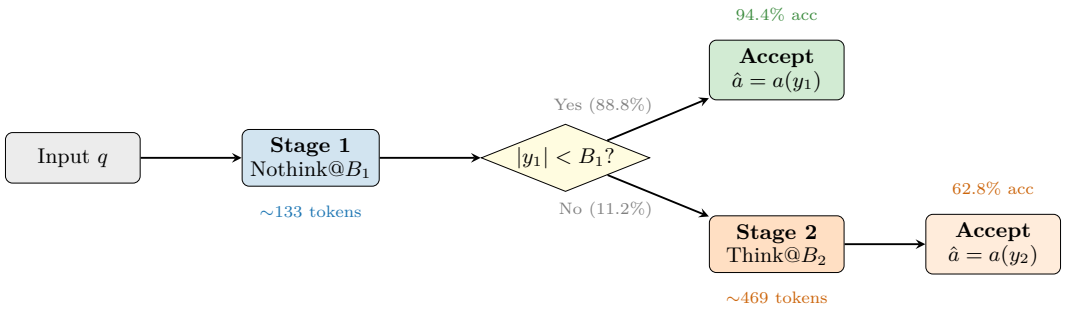

Figure 5: **TOWN inference pipeline.** Stage 1 probes with non-thinking mode at budget B_1. If the model stops early (88.8% of GSM8K), the answer is accepted at ∼133 tokens (94.4% accuracy). Otherwise, Stage 2 routes to thinking mode at budget B_2, recovering additional correct answers on hard problems. Overall: **90.9%** accuracy at **199** average tokens on the full test set (n=1,319).

N Full Theoretical Analysis

The main text presents the core decomposition (Eqs. 1–3) with derivations for the accuracy decomposition and crossover budget. Here we provide the full formal definitions, additional propositions, and remaining proofs.

N.1 Formal Definitions

Definition 1 (Thinking chain length)**.** *For a model $\mathcal{M}$ and question q, let $L(q) \in \mathbb{N}$ denote the* natural chain length*—the total number of output tokens the model would generate in thinking mode if unconstrained (i.e., $b \to \infty$). Let $F_L(t) \triangleq \Pr(L \leq t)$ denote the CDF of L over the question distribution $q \sim \mathcal{Q}$.*

Definition 2 (Truncation rate)**.** *Given token budget b, the* truncation rate *is $\rho(b) \triangleq 1 - F_L(b) = \Pr(L > b)$.*

Assumption 1 (Binary outcome structure)**.** *When the chain completes naturally ($L \leq b$), the model produces a well-formed answer with accuracy $\alpha_c \triangleq \Pr(\textit{correct} \mid L \leq b) \in (0, 1]$. When truncated ($L > b$), the residual accuracy is $\alpha_t \triangleq \Pr(\textit{correct} \mid L > b) \ll \alpha_c$.*

At b=512 on GSM8K (n=1,319), $\alpha_c = 96.3\%$ among natural-stop samples vs. $\alpha_t = 46.6\%$ among truncated samples (Finding 2). At lower budgets, α_t approaches zero.

Remark 1 (Budget-dependent α_t). In practice, α_t varies with b: at b=128, nearly all truncated samples lack any answer ($\alpha_t \approx 0$); at b=512, answer-extraction heuristics and projection passes recover partial answers ($\alpha_t \approx 46.6\%$). The $\alpha_t = 0$ simplification serves as a tight first-order approximation at low budgets.

N.2 Budget Multiplier

Corollary 1 (Budget multiplier)**.** *The budget multiplier $\gamma \triangleq b^*/b_{\text{sat}}$ quantifies the token overhead for thinking to become viable. On GSM8K: $\gamma \approx 2048/512 = 4\times$. On MATH-500: $\gamma > 2048/1024 = 2\times$, since thinking has not yet caught non-thinking at b=2048, confirming that harder tasks demand larger multipliers.*

N.3 Natural Stop as a Confidence Oracle

Proposition 2 (Oracle precision)**.** *The natural-stop event $\mathcal{S}(b) \triangleq \{|y| < b\}$ has positive predictive value $\text{PPV}(\mathcal{S}) = \Pr(\text{correct} \mid \mathcal{S}(b)) = \alpha_c$. For non-thinking mode, $\text{PPV}(\mathcal{S}_{\text{nt}}) \geq \alpha_c$, since early completion selects for easier questions.*

By Hoeffding's inequality, with n=475 natural-stop samples (8B, b=512) and $\hat{\alpha}_c = 0.963$: the 95% lower confidence bound is $\alpha_c \geq 0.919$.

Table 19: **Natural stop as a free confidence oracle.** When a thinking model naturally terminates before exhausting its token budget, the resulting answer is correct with remarkably high probability. This phenomenon is consistent across models and benchmarks. *NatStop*: fraction of samples that stop early (<95% of budget used). Acc_{nat}: accuracy among early-stop samples. Acc_{hit}: accuracy among budget-exhausting samples. Qwen3-8B overall accuracies here include a projection pass (Appendix A); at low budgets this recovers additional answers, yielding higher Acc than Table 1 (which omits projection). The oracle signal (Acc_{nat} vs. Acc_{hit}) is unaffected by this difference. DeepSeek-R1 Acc splits are omitted due to small sample sizes (n=40–80).

Model / Benchmark	**Bud.**	**Acc**	**NatStop**	$\mathbf{Acc_{nat}}$	$\mathbf{Acc_{hit}}$	**Gap**
Qwen3-8B / GSM8K (n=1319)	128	11.8	0.0	—	11.8	—
	256	34.8	1.4	89.5	34.0	+55.5
	512	65.2	37.4	96.3	46.6	**+49.7**
DeepSeek-R1-8B / GSM8K (n=40)	256	22.5	15.0	—	—	—
	512	55.0	85.0	—	—	—
	1024	65.0	97.5	—	—	—
DeepSeek-R1-8B / MATH500 (n=80)	1024	25.0	31.2	—	—	—
	2048	38.7	55.0	—	—	—
	4096	46.2	76.2	—	—	—

Remark 2 (Information-theoretic interpretation). Early stopping implies the model's conditional entropy $H(A \mid y_{1:t^*}) \approx 0$ at $t^* \ll b$—the model is near-certain of its answer. Unlike logit-based confidence, this signal is purely behavioral.

N.4 Inverse Scaling with Model Size

Proposition 3 (Inverse scaling of the thinking tax)**.** *Let L_M denote the chain-length distribution for a model of size M. If L_{M_2} stochastically dominates L_{M_1} for $M_2 > M_1$ (i.e., $F_{L_{M_2}}(b) \leq F_{L_{M_1}}(b)$ for all b), and non-thinking accuracy is approximately size-invariant, then for any fixed budget b:* $\text{Tax}(M_2, b) \geq \text{Tax}(M_1, b)$.

Proof. From Eq. 2: $\text{Tax}(M, b) \approx \text{Acc}_{\text{nt}}(b) - F_{L_M}(b) \cdot \alpha_c$. With Acc_{nt} approximately constant across M, and $F_{L_{M_2}}(b) \leq F_{L_{M_1}}(b)$, the second term decreases, increasing the tax. □

Empirical verification. At b=512: 8B natural-stop rate = 37.4%, 27B = 0.7%. The thinking tax for 27B is 95.5% − 18.3% = 77.2 pp vs. 8B's 93.4% − 65.2% = 28.2 pp—a 2.7× amplification.

Corollary 2 (Crossover budget grows with model size)**.** *Since $b^*(M) \approx F^{-1}_{L_M}(\bar{\alpha}_{\text{nt}}/\alpha_c)$ and $F^{-1}_{L_M}$ shifts right with M: $M_2 > M_1 \implies b^*(M_2) \geq b^*(M_1)$. Larger models require proportionally larger budgets for thinking to become cost-effective.*

O Natural Stop Oracle: Full Analysis

Section 3.2 summarizes the natural-stop oracle. Here we present the complete analysis including the detailed table and figure.

At b=512 with Qwen3-8B, 37.4% of samples terminate before consuming 95% of their token budget (Table 19). Among these, accuracy is **96.3%**; among truncated samples, only 46.6%—a 49.7 pp gap. A stricter definition requiring an explicit "Final answer" marker selects only 6.1% but achieves 93.8%. The effect generalizes to DeepSeek-R1-8B (Table 19).

P Routing Baseline Comparison

Table 20 compares TOWN against budget-matched routing baselines, all routing exactly 11.2% of samples to `think@512`. **Random routing** yields only 85.0% (45.6 regrets over-

whelm 12.7 recoveries). **Inverse token-length routing** performs even worse (83.5%). TOWN's signal is equivalent to a **token-length router**—optimal within the class of budget-utilization-based routers. The **oracle** upper bound (95.8%) leaves a 4.9 pp gap reflecting routing regret and missed recoveries.

Table 20: **Routing baseline comparison** on GSM8K (Qwen3-8B, n=1,319). All budget-matched routers use the same 11.2% routing rate as TOWN.

Method	**Acc. (%)**	**Avg Tok.**	**Route %**	**Recov.**	**Regret**	**Net Δ**
NoThink@256	87.5	146	0.0	0	0	0
Think@512	65.2	477	100.0	109	403	−294
Random	85.0±0.5	200±1	11.2	12.7	45.6	−32.9
Inverse Token-Length	83.5	200	11.2	1	54	−53
Token-Length	90.9	199	11.2	64	19	45
Oracle	95.8	182	8.3	109	0	109
TOWN (Ours)	**90.9**	**199**	11.2	64	19	**45**

Q Full MATH-500 Results

Table 5 in the main text shows the MATH-500 thinking tax results. Here we include the extended version with the additional b=128 data point (see also Table 17 for per-mode details).

Table 21: **Extended MATH-500 thinking tax** (n=500, Qwen3-8B). Includes low-budget results. At budget 256, thinking achieves just 4.2% with 0% early stop rate—every response is truncated.

Budget	**Nothink Acc**	**Think Acc**	**Gap (NT−T)**
128	4.6%	—	—
256	16.6%	4.2%	+12.4pp
512	40.6% [36.4, 45.0]	6.2% [4.2, 8.4]	+34.4pp
1024	59.8% [55.6, 64.0]	18.0% [14.6, 21.4]	+41.8pp
2048	64.4% [60.2, 68.6]	44.0% [39.8, 48.4]	+20.4pp

The thinking tax is confirmed and amplified on MATH-500: at budget 1024, the gap is **41.8 pp** [36.4, 47.0] (paired bootstrap)—even larger than GSM8K's 28.2 pp at the same budget. The crossover has not been reached at 2048 tokens (20.4 pp gap), confirming that harder benchmarks push the crossover further right.

R Extended Discussion

Is the thinking tax obvious in hindsight? While the *mechanism* (truncation) may seem unsurprising, three aspects are not. First, the *magnitude*: a 69.5 pp gap at budget 256 on GSM8K and 41.8 pp at budget 1024 on MATH-500 far exceed what format overhead alone would predict—our theoretical analysis (§4) shows that the tax is not merely "truncation is bad" but follows from a specific interaction between chain-length distributions and answer-completion rates. Second, the *inverse scaling with model size* (§3.3, Proposition 3) is counterintuitive: larger models become *worse* at thinking under the same budget, a phenomenon our theory explains through stochastic dominance of chain-length distributions. Third, the *natural-stop oracle* (§3.2) is a novel signal with 96.3% PPV (Proposition 2), not predicted by the truncation narrative alone.

Why hasn't this been studied before? The thinking tax has gone unnoticed because most reasoning evaluations impose generous budgets (4096–32768 tokens)—well above the crossover point—where thinking mode indeed dominates. In practice, however, production deployments routinely operate at 256–1024 token budgets due to latency, cost, and throughput constraints [Snell et al., 2024]. Our contribution is to systematically characterize this under-studied regime and show that the penalty is not marginal but *severe*—rendering thinking mode strictly inferior across all tested conditions below the crossover. The absence of this analysis from the literature reflects a gap between benchmark evaluation practice (unlimited budgets) and deployment reality (constrained budgets).

The format tax vs. the information tax. A key insight from the 27B nothink results (Appendix U) resolves a natural question: is the tax due to model weakness, or format overhead? The answer is definitively format: **27B nothink@512 achieves 95.5%**—the *highest* accuracy of any configuration we tested—while 27B think@512

achieves only 18.3%. The same model, answering the same questions, goes from near-perfect to near-random depending solely on whether it generates a reasoning chain. This confirms that the thinking tax is a *format tax*: the cost of wrapping answers in verbose reasoning traces that cannot fit within practical budgets.

When *should* one think? Our crossover analysis (§6.2) provides a precise answer: thinking mode requires ∼2048 tokens on GSM8K (4× the non-thinking saturation budget) to match non-thinking accuracy; on MATH-500, the crossover demands an even larger budget, reflecting the longer chains harder problems produce. Our theory (Proposition 4) explains this: the crossover budget b^* corresponds to the $(\bar{\alpha}_{\text{nt}}/\alpha_c)$-quantile of the chain-length distribution—approximately the 97th percentile. Below these budgets, non-thinking mode is strictly preferred. TOWN resolves this tension: think only when the early-stop signal indicates a hard problem.

Why not simply use `nothink@512`? Two factors favor TOWN. First, it recovers genuinely hard cases: 64 of 148 routed samples on GSM8K are genuine recoveries, yielding a net gain of 45 samples (§6.1). Second, harder tasks amplify the thinking tax, making routing between modes more valuable in principle; however, end-to-end TOWN gains on MATH-500 remain unmeasured at the budgets tested, so this remains a theoretical prediction rather than an empirical finding (§6.2). The net recovery condition (Eq. 5) provides a formal criterion for when the cascade is beneficial.

Implications for scaling and the 27B limitation. The inverse scaling (Proposition 3) has troubling implications for frontier models. The 27B model achieves 95.5% in non-thinking mode at budget 512—the highest accuracy of any configuration tested—yet plummets to 18.3% in thinking mode at the same budget. As models grow and chains lengthen (Corollary 2), the crossover budget b^* will shift further right, making the thinking tax *worse*, not better.

Importantly, TOWN does not uniformly benefit all model sizes. At 27B scale with B_1=256, B_2=512, TOWN achieves only 62.8%—*below* `nothink@256` (65.1%)—because the thinking fallback is too weak (think@512 = 18.3% on routed samples). The net recovery condition (Eq. 5) fails: more routing regrets than recoveries. This provides a concrete **deployment criterion**: TOWN should only be applied when the fallback budget B_2 is large enough for thinking mode to add value on the routed hard subset. For 27B, this requires $B_2 \geq 1024$ (where think accuracy rises to ∼49%), implying a higher per-query cost for the fallback stage. This scale-dependent asymmetry is a genuine limitation, and suggests that adaptive routing between thinking and non-thinking modes—rather than unconditional thinking—will become increasingly important for frontier models, but the *minimum viable fallback budget* must be calibrated per model size.

Generalization and limitations. Our experiments span three benchmark families: GSM8K, MATH-500, and five BIG-Bench Hard subtasks (n=3,006 total), three Qwen-family model sizes (8B, 9B, 27B), with cross-architecture validation on DeepSeek-R1-Distill-Llama-8B. The tax appears on both easy and hard benchmarks—and is *larger* on the harder one—while the crossover budget shifts predictably with task difficulty. On BIG-Bench Hard (Appendix V), full-scale evaluation (n=1,187) confirms the tax persists in non-mathematical domains (logical deduction, object tracking, date understanding), with a +33.3 pp gap at budget 256 and crossover between 1024–2048. The mechanism (truncation waste when chain length exceeds budget) is model-agnostic in principle, applying whenever (i) typical chain length exceeds the budget and (ii) the answer appears at the chain's end.

We explicitly scope our claims to **Qwen-family open reasoning models with explicit `<think>` mode under fixed output-token budgets on structured reasoning tasks**. The thinking tax may not apply (or may differ significantly) in three settings: (1) tasks where CoT provides qualitatively different value, such as multi-hop retrieval, iterative code generation, or agentic workflows; (2) closed-source models (e.g., OpenAI o1) where the thinking budget is managed internally; and (3) settings with dynamic or unbounded token budgets where the crossover is always exceeded. While auxiliary DeepSeek-R1 experiments (Appendix C) show consistent utilization collapse and natural-stop patterns, we do not have matched nothink baselines for DeepSeek and therefore restrict our primary claims to the Qwen family. TOWN's binary routing signal could be enhanced with soft confidence scores or logit-based uncertainty; we leave this to future work.

S Theory Verification

Table 22 validates the truncation-waste decomposition (Eq. 1) across three dimensions: budgets, model scales, and benchmarks.

The framework's predictions are **consistent with observations** wherever all components (F_L, α_c, α_t) are directly measurable (8B at $b \in \{256, 512\}$); this is expected since the decomposition is an accounting identity when the binary outcome assumption holds. The crossover prediction at the 97th percentile is self-consistent: at b=2048, if $F_L \approx 0.97$ (i.e., 97% of chains complete within budget), then $0.97 \times 0.963 \approx 0.934$, consistent with the observed crossover. The cross-scale predictions correctly capture the qualitative ordering (27B $\ll$ 9B $\ll$ 8B at same budget) and the mechanism (longer chains $\rightarrow$ smaller F_L $\rightarrow$ more truncation waste). At 9B, observed accuracy (11.5%) slightly exceeds the α_t=0 prediction (<5%) because the projection pass (Appendix A) recovers partial answers from near-complete chains.

The key *predictive* insight is that the crossover budget b^* is fully determined by the chain-length distribution and non-thinking saturation accuracy (Eq. 3): for any model where these quantities are known, the crossover can be predicted *without* running the expensive budget sweep that traditionally determines it. For example, estimating F_L from a small pilot run at a single budget suffices to predict whether thinking mode will be beneficial at a target deployment budget.

Held-out prediction tests and limits. To assess the decomposition's *out-of-sample* predictive power (beyond the in-sample accounting identity), we perform two held-out tests. On BBH, estimating α_c=0.950 and α_t=0.275 from b=512 data predicts b=2048 accuracy within **0.8 pp** and b=1024 within 4.3 pp (main text, §4). The larger b=1024 error arises because α_t increases with budget as more truncated samples produce partial answers that happen to be correct.

Table 22: **Theory verification: predicted vs. observed accuracy.** The framework $\mathrm{Acc}(b) = F_L(b) \cdot \alpha_c + (1 - F_L(b)) \cdot \alpha_t$ is verified as an accounting identity at $b \in \{256, 512\}$ for 8B where all components are directly measurable. The crossover budget b^* corresponds to the 97th percentile of chain lengths (Eq. 3): at b=2048, $F_L \approx 0.97$, yielding $0.97 \times 0.963 = 0.934$—consistent with the observed crossover. Cross-scale: the framework explains the 27B collapse ($F_L(512)_{27\mathrm{B}} = 0.007$ vs. 0.374 for 8B).

Config	$F_L(b)$	α_c	α_t	**Predicted**	**Observed**
Directly measured (8B, GSM8K)					
8B, b=256	0.020	0.973	0.164	18.0%	18.0%
8B, b=512	0.374	0.963	0.466	65.2%	65.2%
Crossover prediction (97th percentile)					
8B, b=2048	∼0.97	0.963	—	93.4%	93.4%
Cross-scale prediction					
9B, b=512	∼0.01	—	—	<5%	11.5%
27B, b=512	0.007	—	0.178	18.3%	18.3%
Cross-benchmark (MATH-500, same model)					
8B, b=512	∼0.00	—	0.062	6.2%	6.2%
8B, b=1024	0.002	1.000	0.178	18.0%	18.0%
8B, b=2048	0.178	0.787	0.365	44.0%	44.0%

This reveals a genuine limitation: the decomposition assumes α_c and α_t are approximately constant across budgets, which holds for moderate extrapolation (BBH b=512→2048: α_t shifts from 0.275 to 0.283, yielding 0.8 pp error) but breaks down when the budget gap is large relative to chain-length variance (MATH-500 b=1024→2048: α_t shifts from 0.178 to 0.365, yielding ∼12 pp error). The practical implication is that the crossover heuristic (97th percentile of chain lengths) remains reliable—it depends only on F_L and $\bar{\alpha}_{\mathrm{nt}}$, not on α_t—but point predictions of thinking-mode accuracy at specific budgets degrade when α_t varies substantially across the prediction range.

T TOWN Pareto-Dominance Proof

Full proof of Proposition 1. Let $p = \Pr(\mathcal{S}_{\mathrm{nt}}(B_1))$ be the early-stop rate. Decompose nothink accuracy:

$$\mathrm{Acc}_{\mathrm{nt}}(B_1) = p \cdot \alpha_c^{\mathrm{nt}} + (1-p) \cdot \alpha_{\mathrm{nt}}^{\mathrm{hard}},$$

where $\alpha_{\mathrm{nt}}^{\mathrm{hard}} \triangleq \Pr(\text{correct} \mid \neg\mathcal{S}, \text{nothink})$. TOWN replaces the second term:

$$\mathrm{Acc}_{\mathrm{TOWN}} = p \cdot \alpha_c^{\mathrm{nt}} + (1-p) \cdot \alpha_{\mathrm{th}}^{\mathrm{hard}}.$$

Subtracting:

$$\mathrm{Acc}_{\mathrm{TOWN}} - \mathrm{Acc}_{\mathrm{nt}}(B_1) = (1-p)\left[\alpha_{\mathrm{th}}^{\mathrm{hard}} - \alpha_{\mathrm{nt}}^{\mathrm{hard}}\right]. \tag{6}$$

The net recovery condition (5) ensures the bracket is positive.

For the cost bound, the expected cost is: $\mathbb{E}[T] = p\bar{t}_1 + (1-p)(B_1 + \bar{t}_2)$. Since $\bar{t}_1 < B_1 < B_2$ and $\bar{t}_2 \le B_2$: $\mathbb{E}[T] < p \cdot B_1 + (1-p)(B_1 + B_2) = B_1 + (1-p)B_2 < B_2$, where the last inequality holds because $p > 0$ and $B_1 < B_2$.

Combining: $\mathrm{Acc}_{\mathrm{TOWN}} > \mathrm{Acc}_{\mathrm{nt}}(B_1)$ with $\mathrm{Acc}_{\mathrm{TOWN}} \gg \mathrm{Acc}_{\mathrm{think}}(B_2)$ (since most chains are truncated at B_2), and $\mathbb{E}[T_{\mathrm{TOWN}}] < B_2 \le \mathbb{E}[T_{\mathrm{think}}(B_2)]$— establishing Pareto-dominance.

On GSM8K (n=1,319): $p = 0.888$, $\alpha_c^{\mathrm{nt}} = 0.944$, $\alpha_{\mathrm{nt}}^{\mathrm{hard}} = (87.5\% - 0.888 \times 94.4\%)/0.112 = 33.0\%$, $\alpha_{\mathrm{th}}^{\mathrm{hard}} = 62.8\%$. Net gain: $0.112 \times (62.8\% - 33.0\%) = 3.3\$ pp ≈ 3.4 pp (observed). □

U Qwen3.5-27B: Non-Thinking vs. Thinking

To complete the cross-scale analysis, we present 27B results in *both* modes, confirming that the thinking tax is even more severe at larger scale.

Key observations:

- The 27B model achieves the **highest accuracy** of any tested configuration in non-thinking mode (95.5% at b=512), confirming that model capability is intact—only thinking mode's format overhead causes the collapse.
- At b=512, the 27B thinking-mode natural-stop rate is just **0.7%**, meaning 99.3% of responses are truncated. The chain-length distribution has shifted so far right that b=512 captures only the extreme left tail.
- The tax grows monotonically with budget (6.3 → 57.2 → 77.2 pp) because non-thinking accuracy improves faster than thinking mode can recover from truncation.

Table 23: **Qwen3.5-27B: nothink vs. thinking on full GSM8K** (n=1,319). At b=512, nothink achieves **95.5%**—the highest accuracy of any configuration—while thinking reaches only 18.3%. The thinking tax is **77.2 pp**, **2.7**× larger than the 8B tax (28.2 pp).

Mode	Budget	Accuracy	Avg Tokens	Early Stop	Tax (pp)
nothink	128	9.9%	127	4.2%	—
nothink	256	65.1%	215	54.5%	—
nothink	512	**95.5%**	249	96.9%	—
thinking	128	3.6%	144	0.0%	6.3
thinking	256	7.9%	272	0.0%	57.2
thinking	512	18.3%	528	0.7%	**77.2**

V Cross-Domain Validation: BIG-Bench Hard

To test whether the thinking tax extends beyond mathematical reasoning, we conduct experiments on five subtasks from BIG-Bench Hard (BBH) [Suzgun et al., 2023]: `boolean_expressions` (n=250), `causal_judgement` (n=187), `date_understanding` (n=250), `logical_deduction_five_objects` (n=250), and `tracking_shuffled_objects_three_objects` (n=250), totaling n=1,187 samples. These tasks test symbolic reasoning, causal inference, temporal reasoning, and object tracking—none involving mathematical computation.

Full-scale results (n=1,187). Table 24 reports the complete evaluation across all five subtasks. At budget 256, nothink achieves **49.9%** vs. thinking's **16.6%**—a **+33.3 pp** tax, confirming the phenomenon generalizes beyond mathematical reasoning. At budget 512, the gap remains substantial (+20.6 pp); at budget 1024, it narrows to +1.4 pp; and at budget 2048, thinking surpasses nothink by 11.0 pp as chains become long enough to complete within the budget. The crossover occurs between budgets 1024 and 2048—later than on GSM8K (∼2048) but consistent with the theory's prediction that harder tasks push the crossover right.

Table 24: **Thinking tax on BIG-Bench Hard** (n=1,187, Qwen3-8B, 5 subtasks). The tax is confirmed on non-mathematical tasks: nothink dominates at budgets ≤1024. Thinking mode surpasses nothink only at budget 2048, consistent with the crossover theory.

Budget	Nothink Acc	Think Acc	Gap (NT−T)
256	49.9%	16.6%	+33.3pp
512	66.4%	45.8%	+20.6pp
1024	75.1%	73.6%	+1.4pp
2048	75.1%	86.0%	−11.0pp

Per-task analysis. The thinking tax varies dramatically across task types, revealing which reasoning patterns are most affected by truncation:

- `tracking_shuffled_objects` (n=250): The most extreme tax. At budget 256, nothink achieves **88.8%** vs. thinking's **0.8%**—an **88.0 pp** gap. The task requires tracking object positions through a sequence of swaps; nothink can produce direct answers, while thinking chains are invariably truncated. Even at budget 512, the gap is 36.8 pp (94.8% vs. 58.0%). Thinking surpasses nothink only at budget 2048 (99.2% vs. 94.8%).
- `boolean_expressions` (n=250): Tax of +36.8 pp at budget 256 (85.6% vs. 48.8%). Thinking recovers quickly: at budget 1024 it surpasses nothink (94.8% vs. 85.6%), reaching 98.0% at 2048. Nothink saturates early (85.6% across all budgets), suggesting this task benefits from reasoning when given sufficient budget.
- `causal_judgement` (n=187): Consistent tax at all budgets, from +25.7 pp at 256 to +3.7 pp at 2048. Thinking does not surpass nothink even at the highest budget, suggesting causal reasoning chains are particularly long and prone to truncation.
- `logical_deduction` (n=250): Tax of +23.6 pp at budget 512 (37.6% vs. 14.0%). Both modes improve dramatically with budget, and thinking surpasses nothink at 2048 (86.8% vs. 78.4%), a −8.4 pp reversal.
- `date_understanding` (n=250): Moderate tax at low budgets (+15.6 pp at 256), with thinking surpassing nothink at budget 1024 (69.2% vs. 52.8%, −16.4 pp). Thinking reaches 82.4% at 2048, 30 pp above nothink—the largest thinking *advantage* of any subtask at high budget, demonstrating that temporal reasoning benefits substantially from extended chains when they can complete.

The BBH results confirm three key patterns from the main text: (1) the thinking tax is large at low budgets and driven by truncation; (2) the crossover budget is task-dependent, occurring between 512 and 2048 depending on chain length requirements; (3) nothink saturates earlier than thinking, so above the crossover, thinking mode provides genuine accuracy gains—the tax is about *budget allocation*, not about whether reasoning is valuable.

W Reproducibility

Models.

- Qwen3-8B: `Qwen/Qwen3-8B` from HuggingFace
- Qwen3.5-9B: `Qwen/Qwen3.5-9B` from HuggingFace

- Qwen3.5-27B: `Qwen/Qwen3.5-27B` from HuggingFace
- DeepSeek-R1-8B: `deepseek-ai/DeepSeek-R1-Distill-Llama-8B`

Random seeds. Main-text GSM8K experiments (full test set, n=1,319) use seed 11 with greedy decoding (τ=0). MATH-500 nothink experiments and the new thinking sweep use seed 42. Fairness experiments use seed 42. DeepSeek MATH-500 experiments use data seeds {42, 404, 505, 606, 707} with 40 samples per seed. All seeds and hyperparameters are recorded in the per-experiment JSON output files.

Generation config. All experiments use greedy decoding: `temperature=0`, `top_p=1.0`, `do_sample=False`. Thinking mode is enabled via `enable_thinking=True` with the model's native `<think>` block format; non-thinking mode sets `enable_thinking=False`. The `max_new_tokens` parameter controls the total output budget equally for both modes. The projection pass (32 extra tokens with thinking disabled) is applied only when thinking mode exhausts its budget without a natural stop.

Software. `transformers` 5.5.0, PyTorch 2.10, CUDA 12.8, Python 3.10. All inference is single-GPU (A100-80GB) with `float16` precision.

Compute. Total compute budget: approximately 200 A100-GPU-hours across all experiments reported in this paper.

NeurIPS Paper Checklist

1. **Claims**
   (a) Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope? [Yes]All accuracy and token-cost claims are derived from reproducible experiments with fixed seeds (Appendix W). The thinking tax is demonstrated across three model sizes (8B, 9B, 27B) within the Qwen3 family, with auxiliary validation on DeepSeek-R1, and two benchmarks (GSM8K, MATH-500). TOWN evaluation uses per-sample matched data (Table 4). Claims about generality are scoped to the validated models and benchmarks (§7).
2. **Limitations**
   (a) Does the paper discuss the limitations of the work performed by the authors? [Yes]The conclusion (§7) discusses limitations including the focus on mathematical reasoning benchmarks, the binary routing signal (vs. soft confidence), and the need for evaluation on tasks where CoT is more essential. The extended discussion in Appendix R further discusses scope limitations.
3. **Theory**
   (a) Does the paper include theoretical results? [Yes]Section 4 presents an analytical framework. The truncation-waste decomposition (Eq. 1) expresses thinking-mode accuracy as a function of the chain-length CDF and conditional accuracies; Proposition 1 shows TOWN achieves a Pareto improvement over fixed-mode strategies under a net recovery condition. The crossover budget formula (Eq. 3) and inverse-scaling analysis (Appendix N) complete the framework. All derivations are in the main text and Appendix T.
4. **Experiments**
   (a) Does the paper include experiments? [Yes]Sections 3 and 6 report comprehensive experiments across GSM8K (full set, n=1,319) and MATH-500, with the Qwen3 family (primary) and auxiliary DeepSeek-R1 validation.
   (b) Does the paper report error bars or confidence intervals? [Yes]We report 95% bootstrap confidence intervals (10,000 resamples) for key comparisons in Tables 4 and 5. Since we use deterministic (greedy) decoding on fixed test sets, variance comes from the test set rather than stochastic generation. Cross-seed validation for DeepSeek-R1 uses 5 data seeds (Appendix C).
   (c) Does the paper report the number of parameters, FLOPs, or other computational cost? [Yes]Average token counts are reported in all tables. Model sizes (8B, 9B, 27B) are specified throughout.
   (d) Does the paper report the computational cost of the experiments? [Yes]Appendix W reports approximately 200 A100-GPU-hours total compute.
5. **Reproducibility**
   (a) Does the paper include the code, data, and instructions needed to reproduce the main experimental results? [Yes]Appendix A provides complete hardware, software, budget control mechanism, and answer extraction pipeline details. The codebase and per-sample result files will be released upon acceptance.
   (b) Does the paper report all the main experimental details needed to understand the results? [Yes]Budget control via `max_new_tokens`, greedy decoding (τ=0), answer extraction pipeline, dataset splits, and random seeds are fully specified in Appendix A.
6. **Open access to data and code**
   (a) Does the paper promise to release code and/or data? [Yes]We will release the complete codebase, per-sample result CSVs, and TOWN implementation scripts.
7. **Experimental Setting/Details**
   (a) Does the paper specify the computing infrastructure used for running the experiments? [Yes]NVIDIA A100-80GB GPUs, PyTorch 2.4.1 (CUDA 12.4), HuggingFace Transformers 5.5.0, bfloat16 precision (Appendix A).

(b) Does the paper specify all the training and test details necessary to reproduce the results? [Yes]No training is performed—TOWN is a training-free inference method. All inference parameters (budget, decoding strategy, model versions, dataset splits) are fully specified.

8. **Broader Impacts**

(a) Does the paper discuss potential broader impacts of their work? [Yes]The conclusion discusses practical implications for LLM deployment: defaulting to non-thinking mode at constrained budgets can simultaneously improve accuracy and reduce computational cost, benefiting both providers and users. Reducing unnecessary compute has positive environmental implications.

9. **Safeguards**

(a) Does the paper describe safeguards for responsible use? [N/A]Our work optimizes inference efficiency for standard reasoning benchmarks and does not introduce new capabilities requiring specific safeguards.

10. **Licenses**

(a) Are the creators of assets used in the paper properly credited? [Yes]GSM8K [Cobbe et al., 2021], MATH [Hendrycks et al., 2021], Qwen3 [Qwen Team, 2025], and DeepSeek-R1 [Guo et al., 2025] are properly cited. All models are publicly available under their respective licenses.

11. **New Assets**

(a) Are new assets introduced in the paper well documented? [Yes]The TOWN inference scripts and per-sample result files will be released with documentation.

12. **Human Subjects**

(a) Does the paper involve crowdsourcing or human subjects? [No]